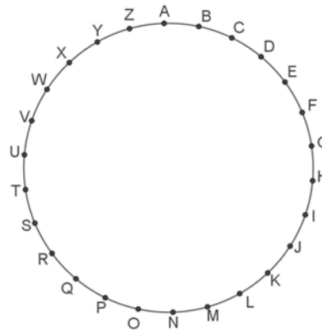


Question 2 – Stage 1



The 26 letters of the alphabet are equally spaced, in alphabetical order, around this circle.

The shortest route along the circle from R to I is 9 steps (anticlockwise).

The shortest route from I to T is 11 steps (clockwise).

To spell RITANGLE in this way, starting at the first letter and always using the shortest route between each pair of successive letters, requires $9+11+7+13+7+5+7=59$ steps. Call this the path length of RITANGLE.

The path length of INTEGRAL is 55 steps (and every step is clockwise!)

Consider the path length for each of the $8!$ arrangements of the letters R-I-T-A-N-G-L-E.

If the shortest path length is x steps and the longest path length is y steps, what is $x \times y$?

Working

```
In [1]: from itertools import combinations, permutations

ritangleLetters = ["R", "I", "T", "A", "N", "G", "L", "E"]
Alphabet = ["A", "B", "C", "D", "E", "F", "G", "H", "I", "J", "K", "L", "M", "N", "O", "P", "Q", "R", "S", "T", "U", "V", "W", "X", "Y", "Z"]

letterIndexes = {}
pairDistances = {}
wordDistances = []

# Find indexes for Letters in "RITANGLE"
for letter in ritangleLetters:
    letterIndexes.update({letter: Alphabet.index(letter)})

# Find Letter Pair distances
for letterPair in combinations(ritangleLetters, 2):
    letterPair = "".join(sorted(list(letterPair)))
    letterOneWayDistance = abs(letterIndexes[letterPair[0]] - letterIndexes[letterPair[1]])
    letterShortDistance = letterOneWayDistance if letterOneWayDistance <= 13 else 26 - letterOneWayDistance # If Longer than half of the circle, use shorter value.
    pairDistances.update({letterPair: letterShortDistance})

# Try each arrangement of "RITANGLE"
for arrangement in permutations(ritangleLetters):
    tempDistanceTotal = 0
    for pair in zip(arrangement[:len(arrangement)], arrangement[1:]):
        tempDistanceTotal += pairDistances["".join(sorted(pair))]
    wordDistances.append(tempDistanceTotal)

wordDistances.sort()
min = wordDistances[0]
max = wordDistances[-1]

print(f"min: {min}")
print(f"max: {max}")
print(f"min*max: {min*max}")

min: 19
max: 81
min*max: 1539
```